

1 Field Types

Overview

HYDRA menu	System administration → System settings → Type definition
FEDRA menu	System administration → System settings → Type definition
Transaction code	ufd
Function authorization	ufd

Purpose

You use type definitions (field type definitions) to display the different standard fields and user fields on the terminal (AIP) and to display the users fields on the MOC.

In addition to the standard field types provided by the system that must NEITHER be changed NOR deleted by the user, the user is able to define separate field types that may, for example, be used for user fields of the operation or the machine within the "U" namespace that has been designed for this purpose.



Field types used to display user fields on the MOC, must not be used on the terminal (AIP). Create new customer-specific field types in the system that you use on the terminal.

For further information on the configuration of user field keys and user fields, refer to the documentation [Configuration_Userfields.pdf](#).

Selection criteria

The application provides the following selection criteria:

Type

Selects the type of the type definition.

Field descriptions

The tabs "General" and "Terminal" provide the fields that can be used for field type definition. The parameters and client used are specified via the *Usage* option of the relevant fields.

General tab**Type**

Unique name of the field type. Special characters and blanks must not be included.

Customized field types must be created with "U:". Standard field types must neither be changed nor deleted.

Syntactic type

A syntactic type is defined. The parameters defined in the syntactic type are used for displaying the single user fields if parameters are missing for the field type. The list described in the section "integration of field types in MOC" or the MPDV Repository indicate which syntactic types are available.

Usage:

- *MOC user fields*

Designation

Self-explanatory name of the field type (need not be unique). If the field type is used in the dynamic dialogs of the terminal, the designation is also used as field label. But it is only used if a designation is not stored in the dynamic dialog.

Usage:

- *MOC user fields*
- *Terminal fields*

Description

Description of the field type (free text field).

Please note for generated, customized MOC applications: When generating an application, this field is used as category of the grid columns of the new application, provided that a user field has been selected and the field type is defined.

Usage:

- *Category of MOC user fields*

Output format

Specifies how the field is displayed in the MOC applications and which characters are shown. The below-mentioned output formats are used by default:

Output format	Description	Example
{0:mpdv_timespan}		
{0:mpdv_timestamp}		
{0:mpdv_timespan_short}		
{0:mpdv_timespan_minutes}		

{0:mpdv_time}		
{0:mpdv_time_short}		
f3	Decimal value with 3 decimal places	
n0	Integer value with 0 decimal places (the decimal places can be defined by replacing "0" by a number)	
HH:mm		
Not applicable	Text field	
Resource configuration	Short date field in the country-specific format	01/07/2013
D	Long date field in the country-specific format	01 July 2013
~-0.0000	To collect negative decimal values, define the output format as follows: "~-0.0000".	

Usage:

- MOC user fields

Input format

Specifies how the field is displayed in the MOC editing dialog and which characters can be entered. The definitions are made in a regular expression subject to the country. For example, the separators used for decimal values are presented differently in different countries.

Please find below some examples for definitions:

Regular expression	Components	Meaning	Example
-?[0-9]{0,9}\R.?[0-9]{0,3}	-?	Negative and positive values are possible	-999999999.999
	[0-9]{0,9}	[Number range from 0-9] {number of digits from 0-9} e.g. 0 to 999999999	
	\R.	Masking \ Country-specific decimal separator	

Regular expression	Components	Meaning	Example
		R.	
	?[0-9]{0.3}	Decimal places optionally ranging between the numbers 0-9 and from 0-3 decimal places	
[0-9]*		Integer, numeric value of an undefined length	12345678900
[0-9]{0.2}		Integer, numeric value ranging between the numbers 0-9 with 0-2 decimal places	99
[^*]{0.250}		Integer value ranging from 0 – 250. Wildcard characters are not supported.	234

Usage:

- MOC user fields

Length

Field length for displaying user fields on the MOC and for displaying the fields in the dynamic dialogs on the Windows terminal for which an existing type definition is stored for the input type.

Usage:

- MOC user fields
- Terminal fields



If an input type has been defined for a field in the dynamic dialogs on the terminal (AIP), the field length defined in the field types is always used. Any deviating definitions that were directly made in the field definition for the dynamic dialog are not used.

Unit

Unit for displaying user fields on MOC and for displaying the fields within the dynamic dialogs at the Windows terminal for which an existing type has been defined for the input type.

Usage:

- MOC user fields
- Terminal fields



If an input type has been defined for a field in the dynamic dialogs on the terminal (AIP), the unit of the field defined in the field types is always used. Any deviating definitions that

were directly made in the field definition for the dynamic dialog are not used.

Terminal index tab

Data type

Specifies the data type used to display contents on the terminal (AIP). The following possibilities are available and transferred to the terminal as described below:

Type definition	Transfer to the terminal
Numeric	Numeric
Text field	ALPHA
Decimal value	FLIESS ("flow")
Checkbox	ALPHA (not used)
Date	DATUM ("date")
Time	ZEIT
Tri-state checkbox	ALPHA (not used)

Usage:

- *Terminal fields*



Data type for MOC user fields

The data type has to be configured in this field for all field types that are used for the MOC user fields!



Note for data type "decimal value"

The syntactic type "userfielddecimal" has to be used for all field types used for MOC user fields and that are configured as data type "decimal value". The input and output formats required for the data type "decimal value" are also defined here. The field type of the terminal must not be used in this case!



Negative decimal values

To collect negative decimal values, define the output format as follows: "~0.0000".

Allowed characters

Defines the allowed characters that may be entered in the fields within the dynamic dialogs at the Windows terminal for which an existing type definition has been defined at the input type within the configuration of the dynamic dialog fields.

Usage:

- *Terminal fields*



If an input type has been defined for a field in the dynamic dialogs on the terminal (AIP), the allowed characters defined in the field types are always used. Any deviating definitions directly within the field definitions for the dynamic dialog are not used/are ignored.

Default values for input

Defines the default value for the input fields in the dynamic dialogs on the Windows terminal for which an existing type definition has been stored for the input type in the configuration of the dynamic dialog fields.

Usage:

- *Terminal fields*



If an input type has been defined for a field in the dynamic dialogs on the terminal (AIP), the default value defined in the field types is always used. Any deviating definitions that were directly made in the field definition for the dynamic dialog are not used.



If default values are used in a workflow step x (not step 1), they must already exist in the first step (also invisible).

Reason: The default values are initialized when opening the workflow in step 1.

From

Possible minimum value of the field when entered in the dynamic dialogs on the Windows terminal.

Usage:

- *Terminal fields*

To

Possible maximum value of the field when entered in the dynamic dialogs on the Windows terminal.

Usage:

- *Terminal fields*

Fill characters

Not used

formula

Not used

•Multiplier

Not used

Divisor

Not used

Summand

Not used

Subtrahend

Not used

Decimal places

Not used



The decimal places displayed on the terminal (AIP) are defined by the output format specified in the "general" tab.

Integration of MOC field types

When defining user fields that are displayed on the MOC, you must enter a field type specifying the display of the user field in relation to the user field key when you assign the user field to a user field key.

If the definition of a parameter is specified for the field type, e.g. the output format, this specification is used for the MOC display. Further definitions of the same parameter, i.e. in this example for the output format, e.g. in the syntactic type, are overwritten and not used.

If single parameters are not set explicitly for the field type and, as a result, are not defined for the field type, the display information from the lower-level configuration hierarchies is used.

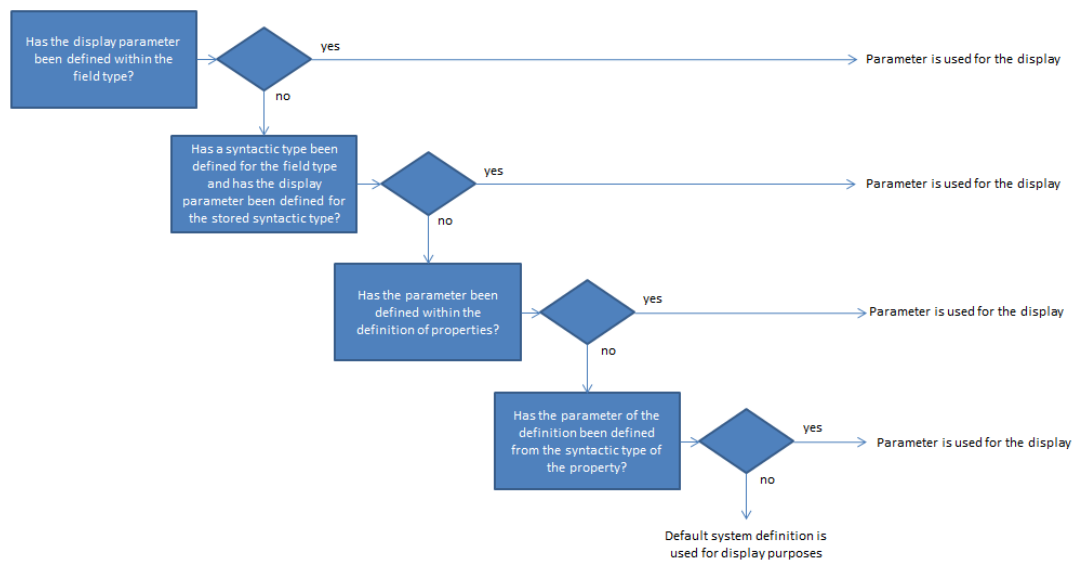
The defined configuration hierarchy is structured as follows:

- 1) Field type definition
- 2) Syntactic type
- 3) Properties with a syntactic type that might be defined (global properties for the fields that are available by default)

A display parameter that is not defined in the field type is determined as follows:

- If parameters are directly defined for the field type they are used for the display.
- If a syntactic type is defined in the field type and not all parameters are set for the field type, the definitions from the syntactic type are used for the missing parameters.
- If no syntactic type is specified for the field type, the parameter defined for the properties is used.
- If no definition is specified in the properties, the definition stored for the syntactic type of the property is used.
- If none of the above-mentioned definitions is available for a parameter, the default system definition is used.

The below diagram shows how the display parameters are identified:



If a syntactic type, which does not exist, is entered in the field type definitions, the required parameter is identified by the system following the process used if the parameter is NOT defined in an existing, syntactic type.

Syntactic types

The following syntactic types are defined for displaying the user fields on the MOC and can be used:

- time
Time field hh:mm:ss
- time_short
Time field hh:mm
- duration
Duration hh:mm:ss
- duration_short
Duration hh:mm
- datetime_date
Date field with date selection
- userfielddatetime
Combined date/time input field
- userfieldstring
Alphanumeric field
- userfieldstringLengthOne
Alphanumeric field of the length 1
- userfieldstringLengthTen
Alphanumeric field of the length 10

- userfieldstringLengthTwenty
Alphanumeric field of the length 20
- userfieldstringLengFourty
Alphanumeric field of the length 40
- userfieldInteger
Integer field
Internal configuration
Output format: n0 (unlimited length)
- userfielddecimal
Decimal field with three decimal places
Internal configuration
Output format: f3
Input format: -?[0-9]{0,9}\R.?[0-9]{0,3}
- userfielddecimal_f1
Decimal field with one decimal place
Internal configuration
Output format: f1
Input format: -?[0-9]{0,9}\R.?[0-9]{0,1}
- userfielddecimal_f2
Decimal field with two decimal places
Internal configuration
Output format: f2
Input format: -?[0-9]{0,9}\R.?[0-9]{0,2}
- userfielddecimal_f3
Decimal field with three decimal places
Internal configuration
Output format: f3
Input format: -?[0-9]{0,9}\R.?[0-9]{0,3}
- userfielddecimal_f4
Decimal field with four decimal places
Internal configuration
Output format: f4
Input format: -?[0-9]{0,9}\R.?[0-9]{0,4}
- userfielddecimal_f5
Decimal field with five decimal places
Internal configuration
Output format: f5
Input format: -?[0-9]{0,9}\R.?[0-9]{0,5}

- `userfielddecimal_f6`
Decimal field with six decimal places
Internal configuration
Output format: `f6`
Input format: `-?[0-9]{0,9}\R.?[0-9]{0,6}`



Please note that syntactic types are case sensitive.



The above-mentioned syntactic types should be used for the configuration of a user field acting as input and output field, since it might not be possible to enter all characters required for the configuration in the "input format" field.

Sample definition for MOC user fields (input and output)

- **Date:**
Create (and assign) a new field type:
- Syntactic type = `datetime_date`
- Output format / input format / length/ unit = empty
- **Time**
Create (and assign) a new field type:
- Syntactic type = `time`
- Output format / input format / length/ unit = empty
- **Duration**
Create (and assign) a new field type:
- Syntactic type = `duration`
- Output format / input format / length / unit = empty
- **Decimal field 2 decimal places**
Create (and assign) a new field type:
- Syntactic type = `userfielddecimal_f2`
- Output format / input format / length / unit = empty
- **Text field (x-digit)**
Create (and assign) a new field type:
- Syntactic type = empty
- Output format = empty
- Input format = optional restriction of the characters to be entered
 e.g. `[0-9a-zA-Z/+#._*%- ]*`
- Length = field length
- Unit = optional unit text

Integration of field types on the terminal

The display and the parameters for the display as well as the values of the single fields can be defined for the terminal (AIP) by specifying a field type for the different fields in the dynamic dialogs on the Windows terminal.

If parameters are not set explicitly for the field type, the configuration, if available, of the dynamic dialogs is used or the INI configuration, e.g. ctaisplay.ini for the terminal.

For the display on the terminal (AIP), the system uses the parameters specified in the type definition in tab "Terminal" and the parameters of the field types in tab "General", e.g. designation.

Special features:

- If a field designation is defined in the dynamic dialog, this designation is displayed. The designation specified for the field type is overwritten.
- With parameters, the definition made for the field types is used and displayed, if available; for example the allowed characters and the default value.